LEARNER GUIDE

Changing the subject, harder cases

Algebraic Expressions and Identities · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

When the subject appears once, undo each step on both sides. When it appears **twice**, no single undo can reach it: gather all its terms on one side and **factorise**, so it appears once — then divide. When it appears as a **square**, undo with a square root; under a **square root**, undo by squaring. Rearranging can create **conditions**: you cannot divide by zero, and a length takes the positive root.

What you are learning to notice

- Every line must say the same thing as the formula you started with — just in a more useful order.
- Collect, factorise, divide: $xy - x = 3y + 2$ becomes $x(y - 1) = 3y + 2$.
- Dividing by $(y - 1)$ needs $y \neq 1$.
- When moving a term across, its sign changes: $-2y$ on one side becomes $+2y$ on the other.
- Check by substituting numbers into the original and the rearranged formula.

Moving through it, one step at a time

If there are fractions, multiply both sides by the denominator. Expand any brackets. Collect every term containing the subject on one side, and everything else on the other. Factorise the subject out, then divide. For a square, take the square root; for a square root, square. State any condition, and check with numbers.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The subject appears twice

Make x the subject of $y = (x + 2)/(x - 3)$.

1. Multiply both sides by $(x - 3)$.	$y(x - 3) = x + 2$
2. Expand the left side.	$xy - 3y = x + 2$
3. Collect the x terms on the left, the rest on the right.	$xy - x = 3y + 2$
4. Factorise: x appears once.	$x(y - 1) = 3y + 2$
5. Divide by $(y - 1)$, which needs $y \neq 1$.	$x = (3y + 2)/(y - 1)$

$x = (3y + 2)/(y - 1)$, for $y \neq 1$.

Example 2 — The subject is squared

Make r the subject of $A = \pi r^2$.

1. Divide both sides by π.	$r^2 = A/\pi$
2. Undo the square with a square root.	$r = \sqrt{(A/\pi)}$
3. r is a radius, a length, so take the positive root.	$r > 0$
4. Check: if $r = 3$, $A = 9\pi$, and $\sqrt{(9\pi/\pi)} = 3$.	✓

***r* = √(*A*/π).**

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (changing the subject when it appears once, and taking out a common factor), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.